1. For the shaded area shown in Fig. 5.2,
   a. Determine centroid of the composite area.
   b. Determine moment of inertia of shaded area about the origin.
   c. Determine moment of inertia of shaded area about an axis that passes through centroid.

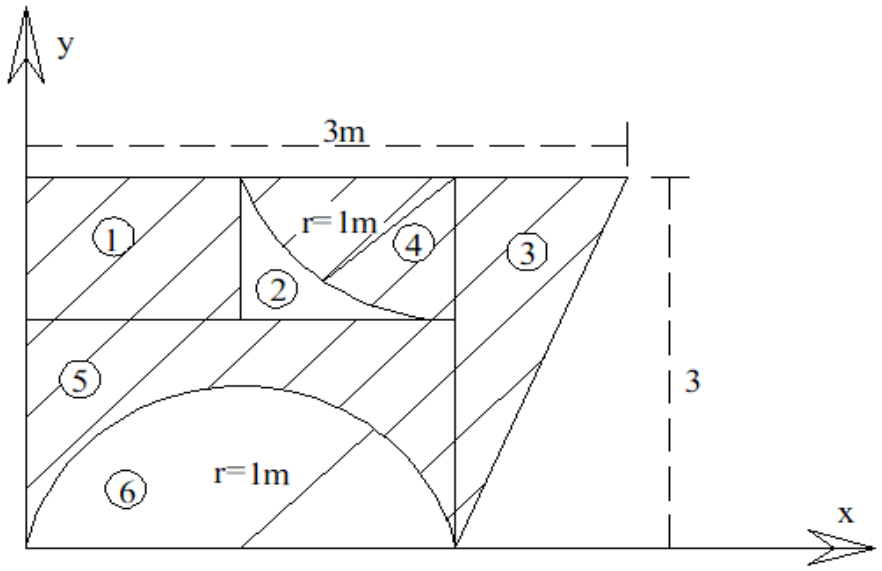


Figure - 5. 1: shaded area

Solution

Step -1: decompose the composite area into known simple area.

i. Rectangle -1 (+ve)
ii. Rectangle -2 (-ve)
iii. Triangle (+ve)
iv. Quarter circle (+ve)
v. Rectangle – 3
vi. Semi – circle (-ve)

i. Rectangle

$A = bh = 1(1) = 1m^2$,

$I_x = \frac{bh^3}{12} = \frac{1(1)}{12} = 0.0833$ $I_y = \frac{hb^3}{12} = \frac{1(1)}{12} = 0.0833$

$x_i = \frac{b}{2} = ½ = 0.5$ $y_i = 2 + \frac{h}{2} = 2 + \frac{1}{2} = 2.5$

$d_x = \frac{b}{2} = ½ = 0.5$ $d_y = 2 + \frac{h}{2} = 2 + \frac{1}{2} = 2.5$

ii. Rectangle (-ve)

$A = -bh = -1(1) = -1m^2$,

$I_x = \frac{-bh^3}{12} = \frac{-1(1)}{12} = -0.0833$ $I_y = \frac{-hb^3}{12} = \frac{-1(1)}{12} = -0.0833$

$x_i = 1 + \frac{b}{2} = 1 + \text{½} = 1.5 \quad y_i = 2 + \frac{h}{2} = 2 + \frac{1}{2} = 2.5$

$d_x = 1 + \frac{b}{2} = 1 + \text{½} = 1.5 \quad d_y = 2 + \frac{h}{2} = 2 + \frac{1}{2} = 2.5$

iii. Triangle (+ve)

$A = \frac{1}{2}bh = \frac{1}{2}1(3) = 1.5m^2$, $I_x = \frac{bh^3}{36} = \frac{1(3)^3}{36} = 0.75 \quad I_y = \frac{hb^3}{36} = \frac{3(1)^3}{36} = 0.0833$

$x_i = 2 + \frac{1}{3}b = 2 + \frac{1}{3}(1) = 2.333 \quad y_i = \frac{2}{3}h = \frac{2}{3}(3) = 2$

$d_x = 2 + \frac{1}{3}b = 2 + \frac{1}{3}(1) = 2.333 \quad d_y = \frac{2}{3}h = \frac{2}{3}(3) = 2$

iv. Quarter circle

$A = \frac{\pi r^2}{4} = \frac{\pi}{4}(1^2) = 0.785m^2$ $I_x = \frac{\pi r^4}{16} - \frac{\pi r^2}{4}\left(\frac{4r}{3\pi}\right)^2 = r^4\left(\frac{\pi}{16} - \frac{4}{9\pi}\right) = 1^4\left(\frac{\pi}{16} - \frac{4}{9\pi}\right) = 0.0547$

$I_y = \frac{\pi r^4}{16} - \frac{\pi r^2}{4}\left(\frac{4r}{3\pi}\right)^2 = r^4\left(\frac{\pi}{16} - \frac{4}{9\pi}\right) = 1^4\left(\frac{\pi}{16} - \frac{4}{9\pi}\right) = 0.0547$

$x_i = 2 - \frac{4r}{3\pi} = 2 - \frac{4(1)}{3\pi} = 1.575 \quad y_i = 3 - \frac{4r}{3\pi} = 3 - \frac{4(1)}{3\pi} = 2.575$

$d_x = 2 - \frac{4r}{3\pi} = 2 - \frac{4(1)}{3\pi} = 1.575 \quad d_y = 3 - \frac{4r}{3\pi} = 3 - \frac{4(1)}{3\pi} = 2.575$

v. Rectangle (+ve)

$A = bh = 2(2) = 4m^2$,

$I_x = \frac{bh^3}{12} = \frac{2(2)^3}{12} = 1.333 \qquad I_y = \frac{hb^3}{12} = \frac{2(2)^3}{12} = 1.333$

$x_i = \frac{b}{2} = \frac{2}{2} = 1 \quad y_i = \frac{h}{2} = \frac{2}{2} = 1$

$d_x = \frac{b}{2} = \frac{2}{2} = 1 \quad d_y = \frac{h}{2} = \frac{2}{2} = 1$

vi. Half circle (-ve)

$A = \frac{-\pi r^2}{2} = \frac{-\pi}{2}(1^2) = -1.57m^2$

$I_x = -\left(\frac{\pi r^4}{8} - \frac{\pi r^2}{2}\left(\frac{4r}{3\pi}\right)^2\right) = -r^4\left(\frac{\pi}{8} - \frac{8}{9\pi}\right) = -1^4\left(\frac{\pi}{8} - \frac{8}{9\pi}\right) = -0.109$

$I_y = \frac{-\pi r^4}{8} = \frac{-\pi}{8}(1)^4 = -0.3925$

$x_i = \frac{D}{2} = r = 1 \quad y_i = \frac{4r}{3\pi} = \frac{4(1)}{3\pi} = 0.425$

$d_x = \frac{D}{2} = r = 1 \quad d_y = \frac{4r}{3\pi} = \frac{4(1)}{3\pi} = 0.425$

a. Centroid of shaded area

Part	$\mathcal{A}$	X_i	y_i	$\mathcal{A}. X_i$	$\mathcal{A}. y_i$
1	1	0.5	2.5	0.5	2.5
2	-1	1.5	2.5	-1.5	-2.5
3	1.5	2.33	2	3.5	3
4	0.785	1.575	2.575	1.24	2.02
5	4	1	1	4	4
6	-1.57	1	0.425	-1.575	-0.67
∑	$A_T = \sum A = 4.71$			$\sum A.\ x_i = 6.165$	$\sum A.\ y_i = 8.35$

$\bar{x} = \frac{\sum A.xi}{A_T} = \frac{6.165}{4.71} = 1.31 \qquad \bar{y} = \frac{\sum A.yi}{A_T} = \frac{8.35}{4.71} = 1.77$

The coordinate of the centroid of the shaded area is $(\bar{x},\bar{y}) = (1.31, 1.77)$

b. Moment of inertia about the origin (indicated axis)

Part	$\mathcal{A}$	$\bar{I}_x$	$\bar{I}_y$	d_x	d_y	$\mathcal{A}d_x^2$	$\mathcal{A}d_y^2$
	1	0.083	0.083	0.5	2.5	0.25	6.25
	-1	-0.083	-0.083	1.5	2.5	-2.25	-6.25
	1.5	0.75	0.083	2.33	2	8.14	6
	0.785	0.0547	0.0547	1.575	2.575	1.95	5.2
	4	1.333	1.333	1	1	4	4
	-1.57	-0.109	-0.3925	1	0.425	-1.57	-0.284

$AT = \sum A = 4.71 \qquad \sum \bar{I}_x = 2.03 \qquad \sum \bar{I}_y = 1.075 \qquad \sum Ad_x^2 = 11.85 \qquad \sum Ad_y^2 = 16.25$

$$I_x = \sum \bar{I}_x + \sum Ad_y^2 = 2.03 + 16.25 = 18.28m^4$$

$$I_y = \sum \bar{I}_y + \sum Ad_x^2 = 1.075 + 11.85 = 12.92m^4$$

$$I_z = I_x + I_y = 18.28 + 12.92 = 31.2m^4$$

Radius of gyration about the origin (indicated axis)

$k_x = \sqrt{\frac{I_x}{A}} = \sqrt{\frac{18.28}{4.71}} = 1.97 \qquad k_y = \sqrt{\frac{I_y}{A}} = \sqrt{\frac{12.92}{4.71}} = 1.656$

c. Moment of inertia about centroidal axis

$\bar{I}_x = \sum I_x - \sum Ad_y^2 \qquad \bar{I}_y = \sum I_y - \sum Ad_x^2 \qquad d_x = \bar{x} = 1.31 \qquad d_y = \bar{y} = 1.77$

$$\bar{I}_x = 18.28 - 4.71(1.77)^2 = 3.52m^4$$

$$\bar{I}_y = \ 12.92 - 4.71(1.31)^2 = 4.84m^4$$

Radius of gyration from the centroidal axis

$\bar{k}_x = \sqrt{\frac{\bar{I}_x}{A}} = \sqrt{\frac{3.52}{4.71}} = 0.864 \qquad \bar{k}_y = \sqrt{\frac{\bar{I}_y}{A}} = \sqrt{\frac{4.84}{4.71}} = 1.014$

Alternatively, $\bar{k}_x = \sqrt{{k_x}^2 - d_y^2} = \sqrt{1.97^2 - 1.77^2} = 0.864$

$$\bar{k}_y = \sqrt{{k_y}^2 - d_x^2} = \sqrt{1.656^2 - 1.31^2} = 1.013$$